

Project 1

Description

- Students may work individually or form a team of up to four members to complete the project. While teamwork is allowed, students are encouraged to complete the project on their own.
- All students must submit their own answers to this project.
- Graduate students are required to answer all questions (i.e., 35 points).
- Undergraduate students may select questions totaling 24 points to answer. The maximum score for an undergraduate student will be capped at 24, regardless of how many questions are attempted.

Submission

- **All** students must complete Project 1, which is listed under the Quiz category.
- **All** students must also submit an **analysis report (using this template)** explaining how the results were derived from the given trace. Students working in the same team may submit an identical report; however, each student must upload the report individually.

Trace Instructions

- Please download the provided pcap trace from the [link](#) to perform the analysis for Project 1.

Relevant IP Addresses

- The trace involves the following IP addresses within the same local area network (LAN):
 - 192.168.1.104
 - 192.168.1.105
 - 192.168.1.1

Attacker Information

- Within the trace, one device (identified by its MAC address) acts as the attacker, launching an ARP spoofing attack.

Question 1 (1 point)

This trace was most likely captured at the switch's mirroring port.

True

False

Analysis:

The capture shows ARP and IP packets exchanged between multiple hosts including traffic that would not normally be visible from a single endpoint. A normal host only sees broadcasts and its own traffic. Seeing conversations between other devices suggests the capture point was observing the LAN broadly.

Question 2 (1 point)

This pcap trace was most likely captured on host 192.168.1.104.

True

False

Analysis:

The trace includes packets exchanged between hosts other than 192.168.1.104. A regular host capture would not see unrelated unicast traffic. This indicates the capture is not limited to that host.

Question 3 (1 point)

192.168.1.104 generates more IP packets than 192.168.1.105 in the given pcap trace.

True

False

Analysis:

Counting IP frames in the trace shows more packets sourced from 192.168.1.104 than from 192.168.1.105. This is consistent with the victim host continuing communication during the spoofing activity.

Question 4 (4 points)

Which of the following IP addresses is likely to be the IP address of the gateway?

Question 4 options:

192.168.1.1

192.168.1.105

192.168.1.104

192.168.1.107

Analysis:

Repeated ARP activity centers around resolving 192.168.1.1 which is typical gateway behavior as many hosts need its MAC address to send off subnet traffic.

Question 5 (4 points)

Select the MAC address (or addresses) that have claimed association with 192.168.1.1.
Note that more than one MAC address may be correct.

Question 5 options:

ff:ff:ff:ff:ff:ff

08:00:27:b8:b7:58

08:00:27:5e:01:7c

08:00:27:2d:f8:5a

Analysis:

The trace shows two ARP replies stating “192.168.1.1 is at ...”. So one is legitimate, the other is spoofed. Broadcast MAC is only used for requests, not identity claims you know.

Question 6 (4 points)

Which MAC address is responsible for launching the ARP spoofing attacks?

Question 6 options:

08:00:27:5e:01:7c

08:00:27:b8:b7:58

08:00:27:2d:f8:5a

ff:ff:ff:ff:ff:ff

Analysis:

The attacker repeatedly sends unsolicited ARP replies mapping the gateway IP to its own MAC. This is a classic ARP poisoning behavior.

Question 7 (4 points)

The attacker attempts to mislead the victim into believing that the attacker's MAC address is associated with a specific IP address. What is that IP address?

Question 7 options:

192.168.1.1

192.168.1.255

192.168.1.105

192.168.1.104

Analysis:

Spoofed replies claim ownership of the gateway address, redirecting victim traffic.

Question 8 (4 points)

Please identify correct statement(s).

Question 8 options:

192.168.1.104 has the MAC address of 08:00:27:b8:b7:58

192.168.1.1 has the MAC address of 08:00:27:2d:f8:5a

192.168.1.1 has the MAC address of 08:00:27:5e:01:7c

192.168.1.105 has the MAC address of 08:00:27:2d:f8:5a

Analysis:

Legitimate ARP exchanges reveal victim host mapping and real gateway mapping whereas spoofed mappings do not change the true associations.

Question 9 (4 points)

Which MAC address (host) is the victim of the attack?

Question 9 options:

ff:ff:ff:ff:ff:ff

08:00:27:2d:f8:5a

08:00:27:5e:01:7c

08:00:27:b8:b7:58

Analysis:

The victim is the host receiving poisoned ARP replies and whose traffic gets redirected.

Question 10 (4 points)

Which IP address is the victim of the ARP spoofing attack?

Question 10 options:

192.168.1.105

192.168.1.10

192.168.1.255

192.168.1.104

Analysis:

Because the MAC resolves to the IP.

Question 11 (4 points)

Please identify correct statement(s).

Question 11 options:

There is no evidence in the trace to show the ARP attack is successful.

The ARP spoofing attack is successful. However, the attacker can only observe i) IP packets from the victim IP to the subnet gateway and ii) IP packets from the victim IP to the external network.

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The ARP spoofing attack is successful. However, the attacker can observe i) IP packets exchanged between the victim IP and the gateway and ii) exchanged between the victim IP and the external network.

Analysis:

After spoofing, traffic continues flowing between victim and gateway through the attacker. This confirms the attacker achieved a man-in-the-middle position and can observe bidirectional communication.